Research Highlights

- Four packet-level attacker actions are modeled as Lie-cone generators on flows.
- Trajectories in the cone provably preserve rate, length, header and MTU invariants.
- FFE cuts perturbation cost 7,500× vs PSO with 2.3× fewer queries.
- Physical feasibility remains 93%–100%, while baseline PFRs drop to 0%–15%.
- Dual defense (MPS + LGAT) slashes FFE success from 51.6% to 5.3% at M=5.